

McKean-Vlasov & Mean-Field Games

Session 21 • The math behind Failure 3 — collective dynamics

Samir Asaf, PhD, CFA, CMA, CTP, CM&AA

Senior Partner, Regent Financial LLC, New York

Former Finance Instructor, Stanford University

Primary Text: Liquidity Illusion (Forthcoming, 2026)

Graduate Finance Course • Spring 2027 • Session 21 of 32

What we'll cover today

1

Why we need a mean-field framework

Recap: McKean-Vlasov externality

2

McKean-Vlasov SDE

Distribution-dependent dynamics

3

Mean-field game equilibrium

Best response $\rightarrow$ fixed point

4

GE-LAV's MFG structure

How the framework instantiates this

5

Solution methods preview

Fixed-point iteration

Mean-field game (MFG): the mathematical structure

An MFG is an equilibrium concept for problems where each agent's optimal decision depends on the distribution of all other agents' states.

Two coupled mathematical objects:

1. HJB EQUATION (PER AGENT)

Each LP solves their own optimal stopping problem (Session 20), but their value function depends on the cross-sectional distribution $\mu(L)$ of other LPs' states.

$V(L, t; \mu) \leftarrow$ each agent's value depends on μ

2. FOKKER-PLANCK EQ (DISTRIBUTION)

Given each LP's best-response decision rule, the cross-sectional distribution $\mu(L, t)$ evolves according to a Fokker-Planck equation.

$\mu(L, t) \leftarrow$ distribution depends on best-response policy

MFG equilibrium = fixed point: agents' best responses $\rightleftharpoons$ resulting distribution

GE-LAV as a mean-field game

GE-LAV instantiates this MFG structure for private capital:

MFG component	GE-LAV instance	Reference
State variable	$L(t)$ — liquidity state experienced by an individual LP	<i>OU process from Session 4</i>
Distribution	$\mu(L, t)$ — cross-sectional distribution of LP states	<i>Continuum of LPs assumed</i>
Control	Exit decision: continue or stop	<i>Optimal stopping (Session 20)</i>
Coupling	Secondary market price $\pi(L, T-t)$ depends on μ	<i>More distressed sellers → higher discounts</i>
Equilibrium	Fixed point where each LP's best response is consistent with μ	<i>Existence proved Track 2 S30</i>

Session 21 summary

What we accomplished today

- 1 Mean-field game (MFG) is the natural framework for problems with cross-sectional coupling
- 2 Two coupled equations: HJB (per agent) + Fokker-Planck (distribution)
- 3 GE-LAV instantiates MFG: state = $L(t)$, control = exit, coupling = secondary price $\pi(L, \mu)$
- 4 Existence of MFG equilibrium for GE-LAV proved in Track 2 Session 30

Next session

Session 22: Fokker-Planck equation in detail — how distributions evolve